

Keyang Xuan

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Education

- University of Texas, Austin**, Ph.D. in Computer Science Aug 2026 –
• Advisor: Yizhong Wang & Jiaxin Pei
- University of Illinois, Urbana-Champaign**, Master in Computer Science Aug 2023 – May 2025
• GPA: 3.95/4.0
• **Selected Coursework:** Advanced NLP, Text Information System, ML System, Data Mining Principle, Language Interface
- University of Minnesota, Twin Cities**, BS in Computer Science; BS in Data Science Sept 2019 – May 2023
• GPA: 3.92/4.0
• **Selected Coursework:** Computer Vision, Deep Learning in NLP, Advanced Machine Learning, Theory of Probability, Applied Statistics, Optimization in Machine Learning

Research Interest

- [R1] Automated Scientific Discovery:** Develop and train large-scale models to accelerate scientific discovery, building the automated research system. (e.g. *ResearchTown*, *TinyScientist*)
- [R2] Social Learning & Modeling:** Design and train agents that exhibit social intelligence and adapt to human interactions. (e.g. *SOTOPIA-RL*, *ResearchTown*, *SocialVeil*)
- [R3] Efficient & Safe AI:** Promote AI democratization through efficient and secure strategies. (e.g. *LEMMA*, *TAMP*)

Publications

- Interactive Evaluation Requires a Design Space** Preprint
*Keyang Xuan**, Peiyang Song*, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, Wenyue Hua, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang, Jiaxin Pei
- Steer2Adapt: Dynamically Composing Steering Vectors Elicits Efficient Adaptation of LLMs** Preprint
Pengrui Han*, Xueqiang Xu*, *Keyang Xuan**, Peiyang Song, Siru Ouyang, Runchu Tian, Yuqing Jiang, Cheng Qian, Pengcheng Jiang, Jiashuo Sun, Junxia Cui, Ming Zhong, Ge Liu, Jiawei Han, Jiaxuan You
- SocialVeil: Probing Social Intelligence of Language Agent under Communication Barriers** Preprint
Keyang Xuan, Pengda Wang, Chongrui Ye, Haofei Yu, Tal August, Jiaxuan You
- SOTOPIA-RL: Reward Design for Social Intelligence** Preprint
Haofei Yu, Zengyang Qi, Yining Zhao, Kolby Nottingham, *Keyang Xuan*, Bodhisattwa Prasad Majumder, Hao Zhu, Paul Pu Liang, Jiaxuan You
- TINYSCIENTIST: An Interactive, Extensible, and Controllable Framework for Building Research Agents** EMNLP 2025 Demo
Haofei Yu*, *Keyang Xuan**, Fenghai Li*, Kunlun Zhu, Zijie Lei, Ziheng Qi, Jiaxun Zhang, Kyle Richardson, Jiaxuan You
- ResearchTown: Simulator of Human Research Community** ICML 2025
Haofei Yu, Zhaocheng Hong, Zirui Cheng, Kunlun Zhu, *Keyang Xuan*, Jinwei Yao, Tao Feng, Jiaxuan You
- APILOT: Navigating Large Language Models to Generate Secure Code by Sidestepping Outdated API Pitfalls** ACSAC 2025
Weiheng Bai, *Keyang Xuan*, Pengxiang Huang, Wu Qiushi, Jianing Wen, Jingjing Wu, Kangjie Lu

TAMP: Token-Adaptive Layerwise Pruning in Multimodal Large Language Models

ACL Findings 2025

Jaewoo Lee, **Keyang Xuan**, Chanakya Ekbote, Sandeep Polisetty, May Fung, Paul Pu Liang

LEMMA: Towards LVLm Enhanced Multimodal Misinformation Detection with External Knowledge Augmentation

KnowledgeLM @ ACL 2024

Keyang Xuan, Yi Li, Fan Yang, Ruochen Wu, Yi R. Fung, Heng Ji

Spatiotemporal Classification with limited labels using Constrained Clustering for large datasets

SDM 2023

Praveen Ravirathinam, Rahul Ghosh, Ke Wang, **Keyang Xuan**, Ankush Khandelwal, Dugan Hilary, Paul Hanson, Vipin Kumar

Open Source Projects

Tiny-Scientist: A Lightweight Framework for Building Research Agents

Lead Developer

 tiny-scientist ★ 127

ResearchTown: Simulator for Human Research Community

Core Maintainer

 research-town ★ 196

Selected Research Experience

University of Illinois, U Lab Advisor: Prof. Jiaxuan You

Sep 2024 – Present

- Proposed Steer2Adapt, a semantic-subspace activation steering framework that composes reusable concept vectors for data-efficient inference-time LLM adaptation.
- Developed cognitive-bias induced communication barrier injected social learning environment for evaluating language agents social interaction capability under communication barriers.
- Developed TinyScientist, a lightweight research agent framework that simulates the full lifecycle of a research process.
- Defined ResearchBench, a benchmark that collects 1000 AI conference paper writing tasks and 200 reviewing tasks to measure the quality of simulations of the research community.

MIT Multisensory Intelligence Lab Advisor: Prof. Paul Pu Liang

Nov 2024 – Feb 2025

- Co-developed TAMP, a token-adaptive pruning framework for multimodal large language models (MLLM) that dynamically determines layer-wise sparsity and prioritizes important multimodal tokens based on diversity and attention-driven activations.
- Achieved an average 1.9% relative performance improvement over the strongest baseline, demonstrating superior performance–sparsity trade-offs compared to standard pruning methods.

University of Illinois, Blender Lab Advisor: Prof. Heng Ji

Dec 2023 – Jun 2024

- Proposed retrieval-augmented based method to improve GPT4V performance on multimodal misinformation detection, achieving a 9% increase on Twitter and a 13% increase on Fakeddit.

University of Minnesota, Security Lab Advisor: Prof. Kangjie Lu

Jun 2023 - Jun 2024

- Designed AST-based method to improve the accuracy of code sanitizer in LLMs and proposed a retrieval-augmented moderation which can achieve a reduction of 89% in vulnerable code recommendations and an average increase of 27.54% in usability.

University of Minnesota, Data Mining Lab Advisor: Prof. Vipin Kumar

Jun 2022 - Jun 2023

- Applied constrained clustering and a hybrid LSTM-CNN architecture to enhance the classification performance on a newly collected remote sensing global waterbody dataset.

Industrial Experience

Research Intern, National Renewable Energy Laboratory

Jan 2024 – Present

- Investigate grid edge resource flexibility prediction with social impact integrated.

- Develop Advanced Non-intrusive Load Disaggregation Method for household energy consumption.

Quantitative Analyst Intern, CITIC Future

Jun 2021 - Aug 2021

- Implemented Double Moving Average Model and rigorously assessing its suitability across diverse commodity futures markets consumption.

Data Analyst Intern, Shanghai Metro Data Tech

May 2021 - Jun 2021

- Crafted SQL queries to precisely extract essential data in alignment with BI product design requirements.

Academia Services

Program Reviewer: ACL Rolling Review 2024, COLM 2026, NeurIPS 2026 Position Track